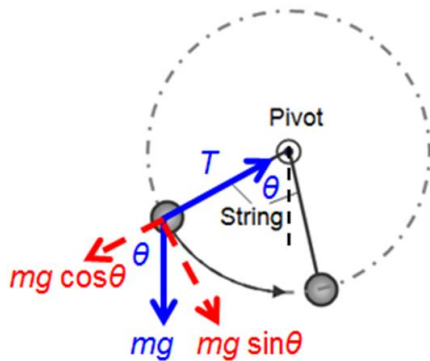


- 8 Centripetal force provided by the resultant force of tension, T and radial component of weight, mg .

Refer to diagram, $T - mg \cos \theta = \frac{mv^2}{R}$

Rearranging, we get

$$T = mg \cos \theta + \frac{mv^2}{R}$$



- 9 Geostationary satellite about Earth has a period of $24 \times 60 \times 60 = 86400$ s.